Luca Petracca

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Summary

AI Engineer with 3 years of experience designing and deploying production AI systems in finance. Specialized in multi-agent LLM architectures that encompass data retrieval, portfolio management, and risk and performance analysis, with end-to-end ownership from architecture to deployment.

Work Experience

AI Engineer Associate *Oct 2025 – Present*
Stifel Europe *Frankfurt, Germany*

- Joined as employee #2 of an AI-focused engineering team, building and shipping production AI systems for financial data at scale as part of a small team in a fast-moving environment.
- Own the core agent layer of a multi-agent Text-to-SQL system using LangGraph: orchestration and planning, schema and example retrieval, SQL and answer generation, and LLM-as-judge evals. Achieved 90% accuracy with RAGAS on curated and synthetic datasets.
- Built hybrid keyword and embedding retrieval with schema-aware ranking, plus row-level access control, auditability, and guardrails for regulated financial workflows.
- Built secure AI solutions with SSO, access controls, dynamic data permissions, and automated guardrails.
- Deployed and maintained scalable AI infrastructure on AWS for reliable use by branch managers, compliance, and financial advisors.

AI Engineer Analyst *Jan 2025 – Oct 2025*
European Central Bank *Frankfurt, Germany*

- Led a Text-to-SQL/RAG product for the ECB Data Portal from proof of concept toward production, later published as an ECB use case.
- Owned AWS data and model infrastructure for an ESCB time-series forecasting hackathon using SageMaker, Glue, S3, IAM, and PySpark.

Data Science Trainee *Feb 2024 – Dec 2024*
European Central Bank *Frankfurt, Germany*

- Built an internal legal RAG prototype on AWS Bedrock to help lawyers draft documents across domestic and European law.
- Built AWS ML pipelines and dashboards tracking the ECB-wide migration of internal data labs from on-premise infrastructure to the cloud.

Software Engineer Intern *Mar 2021 – Sep 2022*
Starting Finance *Rome, Italy*

- Contributed to the development of a mobile app for stock market investment education using Flutter and Dart for Starting Finance, the major financial community for Millennials in Italy with over 1,000,000 followers.

Academic Projects

- **McKinsey & Company - Generative AI in the Law Sector** *Oct 2023 - Dec 2023*
  - Developed a business model focusing on the integration of Gen AI into legal practices
- **Siemens - Methods and Tools for Resilient Industrial IoT** *July 2023 - Aug 2023*
  - Developed a business model for a B2B data-sharing platform in the ESG context for Siemens

Education

EIT Digital - Politecnico di Milano & Universidad Politécnica de Madrid *Sept 2022 - Jul 2024*
Master of Science - Data Science and Artificial Intelligence (GPA 4.0/4.0)

- Minor in Innovation & Entrepreneurship

Università degli Studi di Napoli Federico II *Sept 2018 - Oct 2022*
Bachelor of Science - Computer Engineering

Publications

Petracca, Luca et al., 2025. *Supporting users in seeking data on the ECB Data Portal: a use case for Retrieval Augmented Generation.* European Central Bank, DG Statistics.

Skills

Engineering: Python • TypeScript • SQL • FastAPI • React • PostgreSQL • CI/CD • Docker • Git
AI/ML: Agentic systems • LangGraph • RAG • LLM Evals • Text-to-SQL • Deep Learning • Recommender Systems
Infrastructure: AWS • Terraform • OpenSearch • SageMaker • Bedrock • Azure DevOps • GitHub Actions
Languages: English (IELTS Academic 8.5), Italian (Native), Spanish (Advanced), German (Basic)